Proficient in SOLIDWORKS (80 Hours)				
S.No	**Session**	**Topics**	**Type**	**Duration (Hours)**
1	Session 1&2	**INTRODUCTION TO SOLIDWORKS** **Basic Terminologies:** CAD, CAM, CAE, PDM and Design Process **Concepts Related to CAD:** Design Intent, Feature based modeling, Parametric, Fully associative, Constraints, SWIFT, and Design Process **SolidWorks Graphical User Interface:** Menu bar, Tool bar, Command manager, Configuration manager, Property manager,Feature manager filter, Selection breadcrumbs, Feature manager design tree, Status bar, Heads up view toolbar, Search, Help flyout menu, Task pane, DimXpert & Display manager, Manager Display, Flyout Feature manager design tree, Mouse buttons, Mouse gestures, Accepting features, Context toolbars, Shortcut bars, Keyboard shortcuts, Callouts, Handles,Previews, Pointer feedback, and Selection filters Document options. Customize GUI. **Document Management:** Creating New SOLIDWORKS document, Open existing document, Saving document, and Exit document Terms to Beginning SOLIDWORKS: Origin, Plane, Axis, Face, Edge and Vertex.	Theory/Demo	2.0Hrs
2	Session 3	Practice : Working with Origin, Plane, Axis, Face, Edge & Vertex	Practicals	2.0Hrs
3	Session 4	**SKETCH** **Sketch Elements:** Origin, Plane, Driving dimensions, Driven dimensions, Sketch definitions, Sketch complexity, Where to start a sketch and Sketch snaps. **Sketch Entities:** Inference line, Line, Centerline line, Mid-point line, Circle, Perimeter circle, Arc, Ellipse, Rectangle, Slot, Polygon, Parabola, Spline, Equation driven curve, Point, Text, and grid. **Sketch Relations**: Co-incident, Co-linear, Concentric, Equal, Fix, Merge, Parallel, Perpendicular, Symmetric and Tangent. **Dimensions:** Smart dimension, Instant 2D, Horizontal dimension, Vertical dimension, Locking dimension, Ordinate dimension, Path length dimension, Dimension between two arcs, Using centerline for radial and diameter dimensions, and Fully define sketch.	Theory/Demo	2.0Hrs
4	Session 5	Practice: Drawing a Simple Sketch Profiles	Practicals	2.0Hrs
5	Session 6	**SKETCH EDITING TOOLS** **Sketch tools:** Construction geometry, Fillet, Chamfer, Offset, Convert entities, Trim, Extend, Split, Jog, Mirror, Dynamic Mirror, Move, Copy, Rotate, Scale, Stretch, and Sketch patterns Make path, Close Sketch To Model, Sketch picture, Check Sketch for Feature Sketch Diagnosis, SketchXpert, Rapid Sketch **3D Sketching:** Introduction, Difference between 2D and 3D sketching, Sketch relations in 3D sketches, and Dimensions in 3D sketch	Theory/Demo	2.0Hrs
6	Session 7	Practice: Sketching a Bracket Profile & Wheel Rim Profile	Practicals	2.0Hrs
7	Session 8&9	**PART FEATURES** **Reference Geometry:** Planes, Points, Axes, and Co-ordinates. **Modeling Features:** **Extrude:** From option, Direction1 & 2, Thin features, Draft, and Selected contours. **Revolve:** Axis of revolution, Direction1 & 2, Thin features, and Selected contours **Sweep:** Sketch profile and path, Circular profile, Guide Curves, Profile orientation, Profile twist, Path alignment, Start/End tangency, and Thin feature. **Loft:** Selecting Profiles, Start/End Constraints, Guide curves, Centerline parameters, Sketch tools, and Close loft. Helix and Spiral, Dome **Multibody Concepts:** Introduction, Solid bodies, Mates in multibody, Intersection, Bridging, Combine, Local operations and Exploded views in multi bodies. **Selecting geometries:** Selection context toolbar, Previous selection, Cross selection, Select all, Selection manager,Selection methods and Selection Filter.	Theory/Demo	2.0Hrs
8	Session 10	Practice: Modeling of HUB , Helical Spring • Practice: Modeling a Camshaft with Multibody techniques	Practicals	2.0Hrs
9	Session 11&12	**EDIT FEATURES** Cut Features: Extruded cut, Revolved cut, Swept cut, Lofted cut and Boundary cut. Engineering Features: Fillet, FilletXpert, Chamfer, Shell, Draft, Rib, Wrap, Intersect, Indent, and Flex. Holes: Simple hole, Hole wizard, and Thread. Patterns: Linear pattern, Circular pattern, Sketch driven pattern, Curve driven pattern, Table driven pattern, Fill pattern, Variable pattern, and Mirror. Curves: Split line, Project curve, Composite curve, Curve through XYZ points, and Curve through reference points	Theory/Demo	2.0Hrs
10	Session 13	Practice: Modeling a Extension Spring	Practicals	2.0Hrs
		PART MODELING - ADVANCED TOOLS		

11	Session 15	**Fastening Features:** Mounting boss, snap hook, Snap hook groove, Vent, and Lip/Groove. **Design Library Tools:** Add to library, Add file location, Create new folder, Configure toolbox and Inserting library feature. **Configuration:** Configuration manager, Design table configuration, and Modify configuration dialog box. **Evaluation:** Measure tool, Materials, Calculating Mass Properties, Section properties and Performance evaluation. View types, DimXpert, Equations, 2D to 3D conversions, System options, Trouble shooting	Theory/Demo	2.0Hrs
12	Session 16	Practice: Modeling Wheel Rim with Configuration	Practicals	2.0Hrs
13	Session 17&18	**ASSEMBLY MODELING** Introduction to assembly and assembly environment. **Assembly Approaches:** Bottom up and Top down approach. Insert component **Standard Mates:** Coincident, Parallel, Perpendicular, Tangent, Concentric, Lock, Distance, Angle, and Mate alignment. **Advanced Mates:** Symmetric, Width, Path Mate, Linear/Linear Coupler, and Limit Mate **Mechanical Mates:** Cam, Slot, Gear, Hinge, Rack Pinion, Screw, and Universal Joint Smart mates, Mate reference and Quick mate context toolbar. Sub-assemblies, Flexible sub-assemblies.	Theory/Demo	2.0Hrs
14	Session 19	Practice: Assembly Modeling of Universal Joint	Practicals	2.0Hrs
15	Session 20&21	**ASSEMBLY MANAGEMENT** **Top Down Design:** Creating and editing part in assembly, Layout Sketch **Editing Tools:** Rebuild, Isolate, Move Components, Rotating Components, Collision Detection, Physical Dynamics, Dynamic Clearance, Interference detection, Replace components **Manipulating Components:** Replacing Components, Rotating Components, Move Components, Show/Hidden components, Collision Detection, Physical Dynamics, Dynamic Clearance, Detecting Interference **Creating Pattern:** Assembly all types of component pattern, Mirror Hole series, Hole alignment Exploded views, Bill of Materials in assembly Smart Components, Smart Fasteners, and Working with import and export data	Theoy/Demo	2.0Hrs
16	Session 22	Practice: Assembly Management	Practicals	2.0Hrs
17	Session 23	**SURFACE MODELING** Introduction to surfaces **Surface Modeling Tools:** Boundary surface, Planer surface, Extrude, Revolve, Swept, loft, Offset surface, and Radiate surface **Surface Editing Tools:** Surface cut, Fillet surface, Extend surface, Surface fill, Trim surface, Untrim surface, Mid surface, Move Face, Delete face, Replace Face, Knit surface, Thicken, Thicken cut, and Ruled Surface Freeform	Theory/Demo	2.0Hrs
18	Session 24	Practice: Modelling a Mouse Using Surface Modelling Tools	Practicals	2.0Hrs
19	Session 25&26	**SHEET METAL DESIGN** **Sheet Metal Fundamentals:** Sheet metal design methods, Bend allowance, Bend deduction, and K-factor **Sheet Metal Features:** Base flange, Sketched bend, Edge Flange, Miter Flange, Hem, Jog Creating Break, Corner/Corner Trim, Closed Corners, Rip Inserting Sketched Bend, Fold/Unfold, Forming Tools Inserting Cross Break, Welded Corner Adding Corner Trim, Lofted Trim Extruded cut, Simple hole, Vent Forming tool, Sheet metal gusset Conversion of Solid Body To Sheet Metal Conversion of Imported body to sheet metal Using rip sketches with convert to sheet metal	Theory/Demo	2.0Hrs
20	Session 27	**Practice: Sheet metal- CPU Cover**	Practicals	2.0Hrs
21	Session 28&29	**WELDMENT DESIGN** Introduction to Weldment Group Structural member Trim/Extend End cap, Gusset, Weld bead Extruded cut, Hole wizard, Chamfer Weldment cut list	Theory/Demo	2.0Hrs

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		Sub-weldment, Weldment custom profile		
22	Session 30	Practice: Supporting Frame modeling using weldment	Practicals	2.0Hrs
23	Session 31&32	**DRAWING VIEWS** Getting started with SOLIDWORKS drawings, Projection and views, Create a drawing, Automatic border and working with sheets. **Standard drawing Views:** Standard 3 view, Model view, Exploded view, Model break view, Relative to Model view, Predefined view and Empty view. **Derived drawing Views:** Projected view, Auxiliary View, Detailed view, Crop view, Section view, Broken - Out section, Cut away views, Break view, and Alternate Position view. Drawing properties, Manipulating views	Theory/Demo	2.0Hrs
24	Session 33	Practice: Generating Multiple Drawing Views	Practicals	2.0Hrs
25	Session 34&35	**DIMENSIONS AND ANNOTATIONS** Document properties, Model items **Creating Dimensions:** Smart, Horizontal, Vertical, Baseline, Ordinate, Horizontal Ordinate, Vertical Ordinate, Chamfer, Attach Dimensions, Align Collinear/Radial, Align Parallel/Concentric, Model Dimensions, Auto dimension, DimXpert, Annotations, Spell check **Annotations:** Datum Features, Geometric Tolerance, Surface Finish, Weld symbol, Jog Leaders, Hole Callout, Datum Target, Dowel Pins, Area Hatch, Cosmetic Thread, Balloon, Centre Mark, Centre Lines, Layers, Working With Tables, Bill Of Materials, Hole Table, Sheets And Templates, Sheet Format, Revision symbol, Revision cloud DXF/DWG import and Export	Theory/Demo	2.0Hrs
26	Session 36	Practice: Drafting of Connecting Rod,Creating Bill of Materials Table	Practicals	2.0Hrs
27	Session 37	**MODEL DISPLAY AND RENDERING** Introduction **Model display:** Appearance, Camera, Decal, Scene, Lighting, Walkthrough, Snapshot and Display manager **Rendering:** Rendering with Real view, Ambient occlusion, Rendering with Photoview 360, Caustics, and Final Render.	Theory/Demo	2.0Hrs
28	Session 38	Practice : Rendering of Wheel Rim	Practicals	2.0Hrs
29	Session 39	**DFM CONCEPTS** Rules descriptions DFMXpress SOLIDWORKS Sustainability SOLIDWORKS Costing	Theory/Demo	2.0Hrs
30	Session 40	ASSIGNMENT FOR EVALUATION	Practicals	2.0Hrs